Machine-Learning Wireless Intrusion Detection System (WIDS)

Detecting and blocking Wi-Fi deauthentication flood attacks with a Random Forest

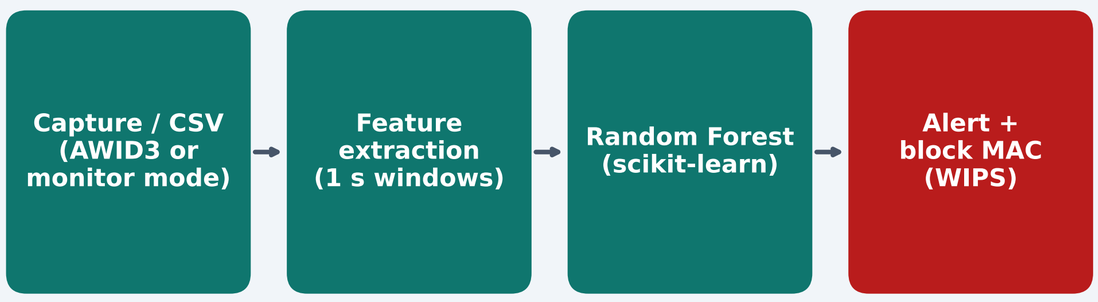
Mawira Victor · 193423 · Mobile and Wireless Security · Strathmore University · 2026

1 The problem

A deauthentication frame is an unauthenticated 802.11 management frame that tells a client to disconnect. An attacker can forge it (spoofing the access point's MAC) and flood the channel, kicking users off repeatedly. It is the first step of evil-twin and handshake-capture attacks, and 802.11w (PMF) is rarely enforced.

Goal: read wireless frames, extract a few simple features, let a classifier label traffic Normal or Malicious, then alert and (WIPS) block the offending MAC.

2 How the system works

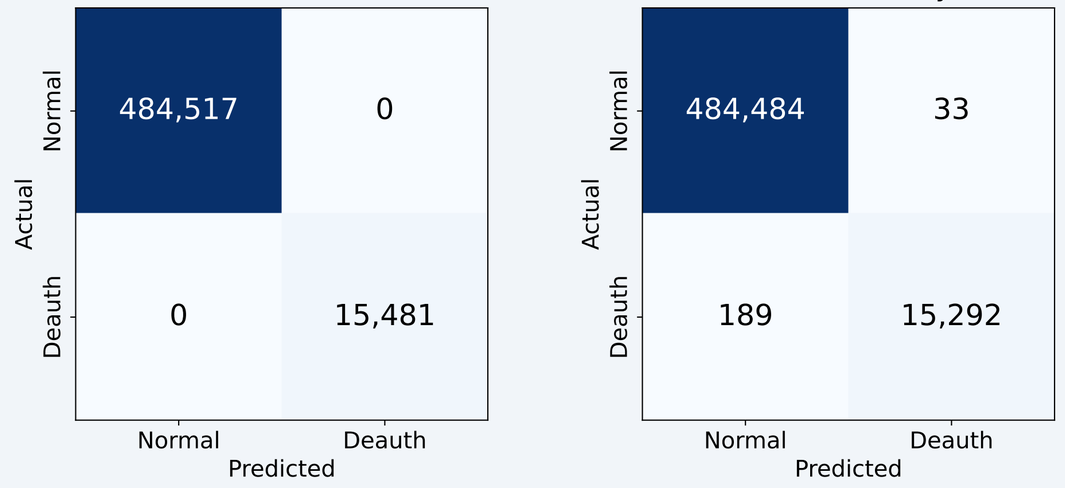


Per frame: type, subtype, is_deauth, length, signal strength (dBm).
Per sender per second: frames/s, deauth frames/s, distinct targets.
Channel-wide: deauth frames/s and deauth ratio.
Split by whole capture files, so the test traffic is unseen.

Replay mode streams a CSV window by window; live mode sniffs a monitor-mode interface (own, authorised lab network only). An alert needs 10 or more malicious frames from one MAC in 1 s.

3 Results

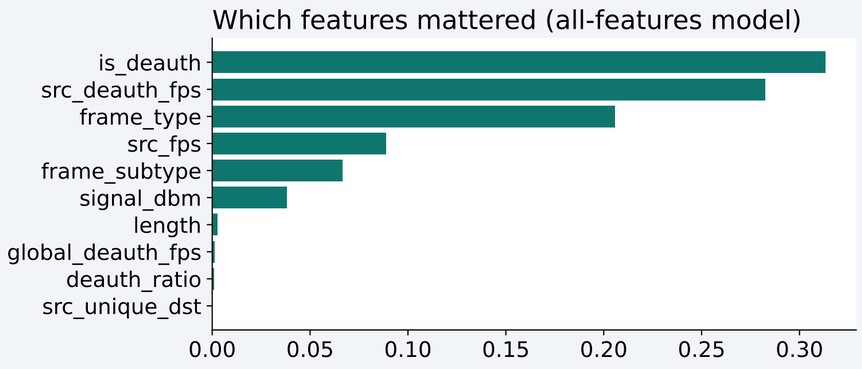
Held-out test: 499,998 frames from 10 of 33 capture files



Behaviour-only model (no subtype / deauth counts)

99.96%	99.78%	98.78%	99.28%
Accuracy	Precision	Recall	F1

Random Forest, 100 trees, depth 12. The simple rule 'subtype = deauth' scores 100.00% accuracy, as good as the all-features model, so the behaviour-only model is the fairer ML result.



4 What I would improve

- A simple rule is as accurate as the full model, so strengthen the behaviour-only features.
- Test on unseen attack types (disassociation, evil twin, KRACK) and other networks.
- Attackers spoof the AP's MAC, so blocking it can block the real AP: detect forged frames with sequence numbers and RSSI.
- Push the blocklist to the AP (hostapd deny list) and recommend 802.11w.
- Measure real-time latency and CPU on a Raspberry Pi sensor.

5 Use of AI coding assistants

I used Claude (Anthropic) to help write the Python code: CSV loading, the 1-second window features, the scikit-learn training and evaluation script, the alert and blocklist logic, and this poster generator. I reviewed and tested the code and can explain each part in the demo: how the features are computed, why the split is by whole files, how the Random Forest votes, and how the WIPS action works.

6 Ethics & scope

All data comes from the public AWID3 dataset (or an isolated lab network I own or am authorised to test). No attack traffic was generated against any third-party network. The WIPS action in this prototype only logs and blocklists a MAC address; it does not transmit frames or jam anything.

Run it:

```
python wids.py train  --data WIDS_project --pattern "Deauth_*.csv"
python wids.py replay --csv Deauth_27.csv --wips
sudo python wids.py live --iface wlan0mon --wips
```